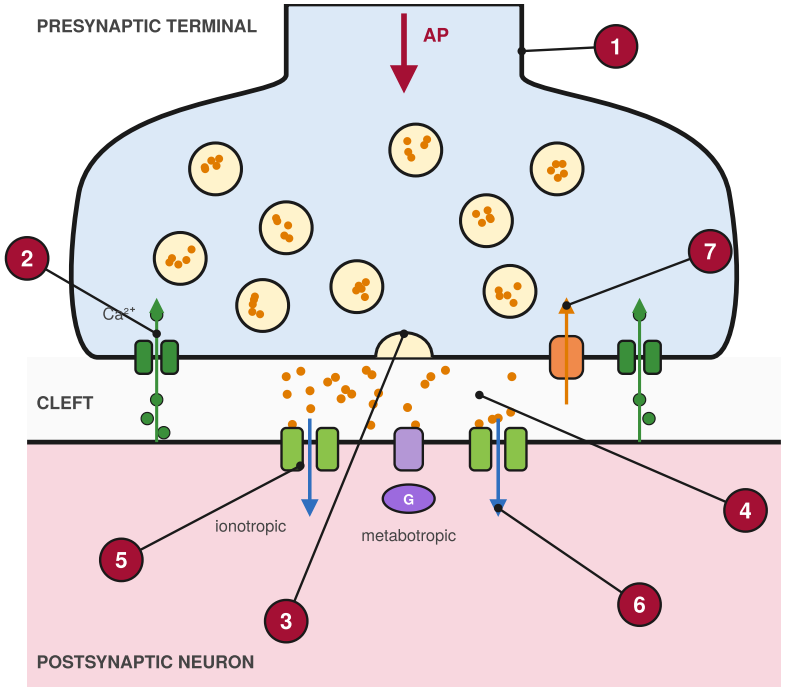
Harvard

Fundamentals of Neuroscience · MicroBachelors® Program

The Synapse & Neurotransmitters

MCB80x · chemical transmission step by step · major transmitters

CHEMICAL SYNAPSE



7 STEPS OF TRANSMISSION

1. **Action potential arrives.** Depolarizes the presynaptic terminal.
2. **Ca^{2+} channels open.** Voltage-gated Ca^{2+} channels let Ca^{2+} in — the key trigger.
3. **Vesicles fuse.** SNARE proteins dock vesicles; exocytosis releases transmitter.
4. **Diffusion.** Transmitter crosses the cleft in well under a millisecond.
5. **Receptor binding. Ionotropic** = ligand-gated channel (fast). **Metabotropic** = G-protein cascade (slow, longer).
6. **Postsynaptic potential. EPSP**: Na^+ in, depolarizes. **IPSP**: Cl^- in or K^+ out, hyperpolarizes.
7. **Clearance.** Reuptake transporters, enzymes (AChE), or diffusion end the signal.

MAJOR NEUROTRANSMITTERS

	Transmitter	Role · exam fact
Glu	**Glutamate** Amino acid · excitatory	Main excitatory transmitter in the CNS; AMPA and NMDA receptors. **FACT** NMDA needs glutamate *and* depolarization (Mg^{2+} block) → LTP.
GABA	**GABA** Amino acid · inhibitory	Main inhibitory transmitter in the brain; $GABA_A$ opens Cl^- channels. **FACT** Benzodiazepines and alcohol enhance $GABA_A$.
Gly	**Glycine** Amino acid · inhibitory	Inhibition in spinal cord and brainstem. **FACT** Strychnine blocks glycine receptors → convulsions.
ACh	**Acetylcholine** Cholinergic	Neuromuscular junction, autonomic system, attention and memory. **FACT** Cleared by AChE. Myasthenia gravis attacks nicotinic ACh receptors.
DA	**Dopamine** Monoamine	Reward, motivation, movement (substantia nigra). **FACT** Parkinson's = DA neuron loss. Cocaine blocks DA reuptake.
5-HT	**Serotonin** Monoamine	Mood, sleep, appetite; raphe nuclei. **FACT** SSRIs (e.g., fluoxetine) block serotonin reuptake.
NE	**Norepinephrine** Monoamine	Arousal, attention, fight-or-flight; locus coeruleus. **FACT** Also a sympathetic nervous system transmitter.

Rules of thumb

- Whether a transmitter excites or inhibits depends on the **receptor**, not the molecule (ACh excites skeletal muscle, slows the heart).
- One AP ≈ one small PSP; the cell fires only when PSPs **summate** to threshold at the hillock.

Personal MCB80x study chart — not an official HarvardX/edX publication.